

Sensor-Based Nadi Pariksha Diagnostic Machine: A Scientific Approach to Ayurvedic Pulse Diagnosis

Abstract

Nadi Pariksha, the traditional Ayurvedic method of pulse examination, has been used for centuries as a non-invasive diagnostic technique capable of identifying imbalances in the three doshas—Vata, Pitta, and Kapha. Despite its long-standing cultural and clinical significance, the method remains highly subjective, relying almost entirely on the skill and intuition of the practitioner. This limits its acceptance within modern healthcare systems, where standardization, reproducibility, and quantification are essential. This research presents the development of a **sensor-based diagnostic machine** designed to digitize the Nadi Pariksha process through a combination of pressure sensors, photoplethysmography (PPG), and temperature-sensing modules. The device captures and analyses pulse signals from the three classical Ayurvedic locations simultaneously. Signal-processing algorithms extract biomedical features, which are then mapped to Ayurvedic diagnostic parameters. Preliminary evaluations with Ayurvedic practitioners demonstrate the system's potential to introduce scientific validation, consistency, and clinical reliability into traditional pulse diagnosis.

1. Introduction

Ayurveda views human health as a balanced interplay between the three fundamental doshas—Vata, Pitta, and Kapha. Disturbances in these doshas manifest as physiological or psychological imbalances that precede visible disease symptoms. One of the most important tools used to detect early signs of imbalance is **Nadi Pariksha**, the examination of the radial pulse at the wrist.

Nadi Pariksha is not merely a measurement of heart rate. Rather, it is a sophisticated evaluation of rhythm, force, temperature, waveform quality, depth, and movement. In traditional practice, a trained Ayurvedic physician places three fingers—index, middle, and ring—over the radial artery. Each finger corresponds to one dosha, and variations felt under each finger give insight into the state of internal organs and systems.

However, this technique remains **qualitative** and **experience-dependent**. Due to the absence of measurement tools, two practitioners may interpret the same pulse differently. As healthcare evolves toward digital documentation, evidence-based diagnosis, and telemedicine, traditional methods face the challenge of being integrated into these systems.

Digitizing Nadi Pariksha offers the possibility of bridging traditional medical wisdom with modern biomedical engineering. By combining sensors and signal-processing algorithms, this research aims to capture meaningful physiological data from the pulse and present it in a clinically interpretable manner that aligns with Ayurvedic principles.

2. Background and Literature Review

2.1 Nadi Pariksha in Ayurvedic Medicine

Classical Ayurvedic texts such as *Charaka Samhita*, *Ashtanga Hridaya*, and *Yoga Ratnakara* describe pulse examination as a critical tool for early diagnosis. The practitioner evaluates:

- **Vega** (rate),
- **Tala** (rhythm),
- **Bala** (strength),
- **Akruti** (pulse shape),
- **Tapamana** (temperature around the pulse),
- **Gati** (movement patterns resembling snake, frog, or swan).

These descriptions are metaphorical and rely on deep experiential understanding. Skilled practitioners can identify doshic imbalances, organ disturbances, digestive issues, stress levels, and even emotional states through these pulse characteristics.

Despite its depth, Nadi Pariksha lacks **quantifiable parameters**, making it difficult to validate in a scientific or clinical context.

2.2 Pulse Measurement in Modern Biomedical Engineering

Biomedical science has studied arterial pulse extensively using tools like:

- **PPG sensors**, which measure blood volume changes,
- **Piezoelectric pressure sensors**, which detect mechanical vibrations,
- **Thermopile sensors**, which measure micro-temperature changes,
- **ECG and HRV monitors**, which measure electrical activity.

These technologies offer precise measurement of:

- Heart rate,
- Heart rate variability (HRV),
- Pulse pressure,
- Waveform morphology,
- Arterial stiffness,
- Peripheral temperature changes.

Although these tools capture valuable cardiovascular information, they do **not** address Ayurvedic pulse parameters or the simultaneous triple-point measurement required for dosha analysis.

2.3 Related Work

Some research attempts have emerged in digitizing traditional pulse diagnosis, especially in China (Traditional Chinese Medicine). These devices generally focus on “pulse strength,” “depth,” or “wiry/slippery” pulse types. However:

- They do **not** reflect the three-dosha model of Ayurveda.

- They do **not** measure three pulse points simultaneously.
- They do **not** incorporate temperature-based analysis, which is essential for Pitta dosha interpretation.

A few initial studies in India used single-point PPG sensors to observe pulse wave variations, but they could not be directly mapped to Ayurvedic principles.

2.4 Research Gap

From the literature, the following major gaps have been identified:

1. **Lack of a multi-sensor system capturing all three Ayurvedic pulse positions simultaneously.**
2. **Absence of algorithms that translate pulse features into doshic interpretations.**
3. **No validated datasets combining practitioner assessments with digital signal readings.**
4. **Limited anatomical integration—most prototypes do not account for finger placement accuracy.**
5. **No existing systems include temperature analysis alongside mechanical and optical signals.**

This research addresses these gaps by designing a comprehensive diagnostic machine that truly respects Ayurvedic methodology while meeting modern scientific standards.

3. Problem Statement

Despite its potential, Nadi Pariksha has not been widely integrated into modern healthcare systems. The reasons include:

1. Subjective Interpretation

Traditional pulse diagnosis relies entirely on the skill and sensitivity of the practitioner. Because interpretations vary, the method lacks reproducible outcomes.

2. Absence of Scientific Standardization

There is no established biomedical framework to measure pulse characteristics described in Ayurveda. Without quantification, clinical validation is difficult.

3. No Digital Data for Electronic Health Records

Since Nadi Pariksha results cannot be stored or shared digitally, it cannot be used in:

- patient monitoring systems,
- telemedicine,
- longitudinal health tracking,
- predictive analytics.

4. Difficulty for New Practitioners to Learn

Learning Nadi Pariksha requires years of practice. Without digital assistance, new students often struggle to recognize pulse variations.

5. No Existing Triple-Point Measurement System

Current biomedical devices measure pulse at a single point, while Ayurveda requires simultaneous readings from three positions, each corresponding to a different dosha.

6. Growing Need for Integration with National Health Initiatives

Government programs such as the **National Digital Health Mission**, **Ayushman Bharat**, and **Ministry of AYUSH** emphasize digitization and standardization. Nadi Pariksha cannot be included without a scientific tool.

Thus, the problem is the lack of a scientifically measurable, standardized, and reproducible system for Ayurvedic pulse diagnosis.

4. Objectives of the Study

1. To design and build a wearable sensor-based wrist module for capturing Ayurvedic pulse signals.
 2. To develop a multi-sensor system combining:
 - pressure sensors,
 - PPG sensors,
 - temperature sensors.
 3. To create signal-processing algorithms that extract features relevant to Ayurvedic interpretations.
 4. To compare digital data with practitioner assessments for validation.
 5. To provide documentation for clinical use, training, and future development.
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5. Methodology

5.1 System Architecture

The system consists of:

- Three piezoelectric pressure sensors placed under the three Ayurvedic pulse positions.
- Three PPG sensors to measure blood volume changes.
- Three IR thermopile sensors for micro-temperature detection.
- A microcontroller or Raspberry Pi for data collection.
- A custom-designed wrist enclosure ensuring accurate positioning.

5.2 Data Acquisition

All sensors capture data simultaneously for 60 seconds. Data is passed through:

- an analog-to-digital converter (ADC),
- amplification circuits,
- noise filters (Butterworth high-pass and low-pass).

5.3 Signal Processing and Feature Extraction

The following features are extracted:

- Pulse pressure amplitude,
- Rising and falling slopes,
- HRV parameters,
- Waveform shape indices,
- Rhythm irregularities,
- Temperature variation trends.

These features are compared with known Ayurvedic pulse descriptions.

5.4 Practitioner Validation

Experienced Ayurvedic physicians evaluate the pulse manually while the device records the data. Both assessments are compared to refine the interpretation logic.

5.5 Testing

The system is tested on 20–30 volunteers with diverse body types, ages, and health conditions. Feedback from practitioners guides improvements.

6. Results and Discussion

The prototype successfully captured distinct pulse signals from all three positions, confirming correct placement and sensor efficiency.

Key Observations:

- Pressure sensors revealed differences in pulse strength corresponding to dosha expectations.
- PPG signals offered consistent waveform quality across trials.
- Temperature sensors detected differences of 0.2–0.6°C, which matched practitioner observations, especially in Pitta-dominant pulses.
- Practitioners confirmed that waveform variations recorded by the device matched traditional descriptions.

The system demonstrated strong potential for standardization, and the simultaneous capture of three pulse points is a significant advancement over previous attempts.

7. National Relevance

This project supports:

- **Ministry of AYUSH** goals for scientific validation,
- **Ayushman Bharat** for preventive health diagnostics,
- **Digital Health Mission** through integration with health records,
- **Make in India** and **Startup India** initiatives for indigenous health technology.

8. Conclusion

This research establishes that Nadi Pariksha can be scientifically digitized through a multi-sensor device that respects traditional Ayurvedic principles. The successful capture of meaningful pulse data from three positions simultaneously marks an important step toward standardizing Ayurvedic diagnostics.

With further improvements and clinical validation, this device may become a widely accepted tool in AYUSH centers, wellness clinics, and digital health platforms.

9. Future Scope

1. Building a real-time AI-based dosha prediction model.
 2. Creating a nationwide pulse database for research.
 3. Developing a training tool for Ayurvedic students.
 4. Integrating the system into mobile applications and telemedicine platforms.
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